

MATH 565 Monte Carlo Methods

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Test 1

Tuesday, September 15, 2026

Instructions:

- i. This test has **FOUR** questions. Attempt all. The maximum number of points is **100**.
- ii. The time allowed is 75 minutes.
- iii. This test is closed book, but you may use **4** double-sided letter-size sheets of notes.
- iv. No calculators or other electronic devices are allowed. Phones must be placed in your bags under your desks or at the front of the room. Hands must be on top of your desks.
- v. You are expected to simplify any expressions that can be simplified as whole numbers, e.g., $36/15 = 12/5$, but you may leave other expressions as is, e.g., $\sqrt{47}$ or $18/29$.
- vi. Show all your work to justify your answers. Answers without adequate justification will not receive credit.
- vii. Print your name clearly on each answer sheet.
- viii. Off-site students may contact the instructor as directed by your syllabus.

ChatGPT/Codex assisted in the construction of this exam.

I understand these instructions and have not relied on any help for this exam beyond what is allowed, nor provided any help to anyone else beyond what is allowed.

Q	Score
1	
2	
3	
4	
Total	

Signature	Date
Printed Name	

Distribution	Sample Space	$q(x)$	μ	σ^2
Bernoulli — Bern(p)	$\{0, 1\}$	$p^x(1-p)^{1-x}$	p	$p(1-p)$

Upper quantiles: z_α satisfies $\mathbb{P}(Z > z_\alpha) = \alpha$ for $Z \sim \text{Norm}(0, 1)$

α	0.995	0.99	0.975	0.95	0.9	0.1	0.05	0.025	0.01	0.005
z_α	-2.58	-2.33	-1.96	-1.64	-1.28	1.28	1.64	1.96	2.33	2.58

The following IID uniform random numbers may be used in the problems:

	U_0	U_1	U_2	U_3
Value	0.64	0.16	0.36	0.25

1. (21 points) Suppose that $Y = f(\mathbf{X})$, where $\mathbf{X} \sim \text{Norm}(\mathbf{0}, \mathbf{I})$. For each estimator of $\mu := \mathbb{E}(Y)$, explain whether it is or is not biased for an arbitrary f . Let $\mathbf{X}_0, \mathbf{X}_1, \dots \stackrel{\text{iid}}{\sim} \text{Norm}(\mathbf{0}, \mathbf{I})$.

a. (7 points) $\hat{\mu} = \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{X}_i)$

Answer: This estimator is unbiased because each $\mathbf{X}_i$ has the same distribution as $\mathbf{X}$. By linearity of expectation,

$$\mathbb{E}(\hat{\mu}) = \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}[f(\mathbf{X}_i)] = \frac{1}{n} \sum_{i=0}^{n-1} \mu = \mu.$$

b. (7 points) $\hat{\mu} = \frac{1}{n} \sum_{i=0}^{n-1} f(-\mathbf{X}_i)$

Answer: This estimator is also unbiased. A zero-mean multivariate normal distribution is centrally symmetric, so $-\mathbf{X}_i \sim \text{Norm}(\mathbf{0}, \mathbf{I})$ and therefore

$$\mathbb{E}(\hat{\mu}) = \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}[f(-\mathbf{X}_i)] = \mu.$$

$-\mathbf{X}_i \sim \text{Norm}(\mathbf{0}, \mathbf{I})$ means $f(-\mathbf{X}_i) \sim f(\mathbf{X}_i)$ but NOT $f(-\mathbf{X}_i) = f(\mathbf{X}_i)$.

c. (7 points) $\hat{\mu} = \frac{1}{n} \sum_{i=0}^{n-1} f(\mathbf{1} + \mathbf{X}_i)$

Answer: This estimator is biased in general. Since $\mathbf{1} + \mathbf{X}_i \sim \text{Norm}(\mathbf{1}, \mathbf{I})$ rather than $\text{Norm}(\mathbf{0}, \mathbf{I})$,

$$\mathbb{E}(\hat{\mu}) = \mathbb{E}[f(\mathbf{1} + \mathbf{X})],$$

which need not equal $\mu = \mathbb{E}[f(\mathbf{X})]$ for an arbitrary f .

$f(\mathbf{1} + \mathbf{X}_i) \neq 1 + f(\mathbf{X}_i)$ in general.

2. (39 points) Suppose that X has unnormalized target density $\varrho_{\text{tar}}(x) = \sqrt{x}$ on $[0, 1]$. Generate a sample X_0 in two different ways using the uniform random numbers above.

Answer: The normalization constant is

$$\int_0^1 \sqrt{x} \, dx = \frac{2}{3},$$

so the normalized density and distribution function are

$$\varrho_{\text{tar}, \text{norm}}(x) = \frac{3}{2}\sqrt{x}, \quad F(x) = \int_0^x \frac{3}{2}\sqrt{t} \, dt = x^{3/2}, \quad 0 \leq x \leq 1.$$

For inverse transformation, $F^{-1}(u) = u^{2/3}$. Using $U_0 = 0.64$ gives

$$X_0 = F^{-1}(U_0) = U_0^{2/3} = 0.64^{2/3} \approx 0.743.$$

Before applying the inverse CDF in the quantile transform, be sure to normalize the target density so that its CDF increases from 0 to 1.

Another valid inverse-transform construction is $X_0 = F^{-1}(1 - U_0)$, since $1 - U_0$ is also uniformly distributed. Here this gives $X_0 = (1 - 0.64)^{2/3} = 0.36^{2/3} \approx 0.506$.

Alternatively, use acceptance-rejection with the unnormalized target density and proposal density $\varrho_{\text{prop}}(x) = 1$ on $[0, 1]$.

For acceptance-rejection, we do not need to normalize the target density.

Since

$$\varrho_{\text{tar}}(x) = \sqrt{x} \leq 1 = \varrho_{\text{prop}}(x),$$

the envelope constant is 1. We propose $X = U_0 = 0.64$ and accept when U_1 is at most $\sqrt{X}$. Because

$$U_1 = 0.16 < \sqrt{0.64} = 0.80,$$

the first proposal is accepted, giving $X_0 = 0.64$.

3. (30 points) A Monte Carlo simulation produces $Y_0, \dots, Y_{899} \stackrel{\text{iid}}{\sim} \text{Bern}(p)$, where $Y_i = 1$ indicates a failure. The observed sample mean failure rate is

$$\hat{p} = \frac{1}{900} \sum_{i=0}^{899} Y_i = 0.100.$$

- a. (20 points) Construct an approximate 99% confidence interval for the population mean failure rate $p = \mathbb{E}(Y_i)$.

Answer: Using the normal approximation and $z_{0.005} = 2.58$, the estimated standard error is

$$\widehat{\text{SE}}(\hat{p}) = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = \sqrt{\frac{0.100(0.900)}{900}} = 0.01.$$

Thus the approximate 99% confidence interval is

$$\begin{aligned}\hat{p} \pm z_{0.005} \widehat{\text{SE}}(\hat{p}) &= 0.100 \pm 2.58(0.01) \\ &= 0.100 \pm 0.0258,\end{aligned}$$

or $[0.0742, 0.1258]$.

The critical value is 2.58, not 2.33, because this is a two-sided 99% confidence interval.

- b. (10 points) Assuming that p is close to the observed sample mean, how many IID samples are required for a 99% confidence interval that is one-tenth as wide?

Answer: For fixed confidence level and failure rate, the interval width is proportional to $1/\sqrt{n}$. Therefore, making the width one-tenth as large requires

$$\frac{1}{\sqrt{n_{\text{new}}}} = \frac{1}{10} \frac{1}{\sqrt{900}},$$

and hence

$$n_{\text{new}} = 10^2(900) = 90,000.$$

4. (10 points) Let Y have finite variance, and let $\hat{\mu}_n = \frac{1}{n} \sum_{i=0}^{n-1} Y_i$ be the sample mean of n IID random variables with the same distribution as Y . Under what condition(s), if any, is $\hat{\mu}_n^2$ an unbiased estimator for $\mathbb{E}(Y^2)$? Explain your answer.

Answer: Since $\hat{\mu}_n$ is unbiased for $\mathbb{E}(Y)$,

$$\mathbb{E}(\hat{\mu}_n^2) = \text{var}(\hat{\mu}_n) + [\mathbb{E}(\hat{\mu}_n)]^2 = \frac{\text{var}(Y)}{n} + [\mathbb{E}(Y)]^2.$$

On the other hand,

$$\mathbb{E}(Y^2) = \text{var}(Y) + [\mathbb{E}(Y)]^2.$$

Therefore, $\hat{\mu}_n^2$ is unbiased for $\mathbb{E}(Y^2)$ if and only if

$$\left(1 - \frac{1}{n}\right) \text{var}(Y) = 0.$$

Thus the condition holds when $n = 1$ or $\text{var}(Y) = 0$. In particular, for $n > 1$, it holds only when Y is constant almost surely.

Test 1 Scores (out of 100)

Number of Students: 20, Minimum: 4, Maximum: 96, Mean: 49, Median: 49

Standard Deviation: 27.5, Quartiles (Q1, Q3): (31.5, 69)

9		6
8		049
7		5
6		123
5		23
4		16
3		567
2		8
1		36
0		48